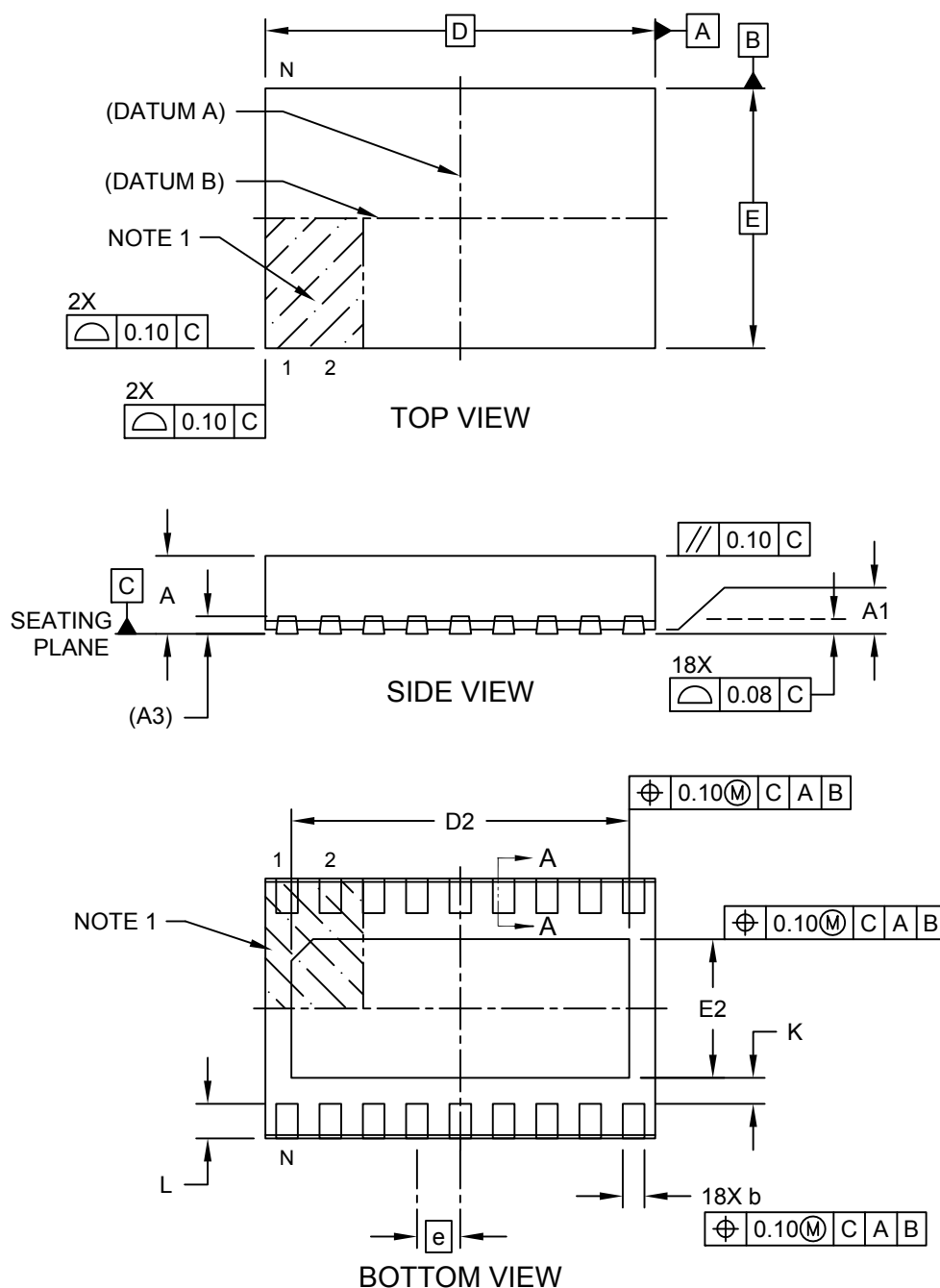


Packaging Diagrams and Parameters

18-Lead Very Thin Dual Flatpack No-Lead Package (QQB) 4.5x3 mm Body (VDFN) With Stepped Wettable Flanks

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>

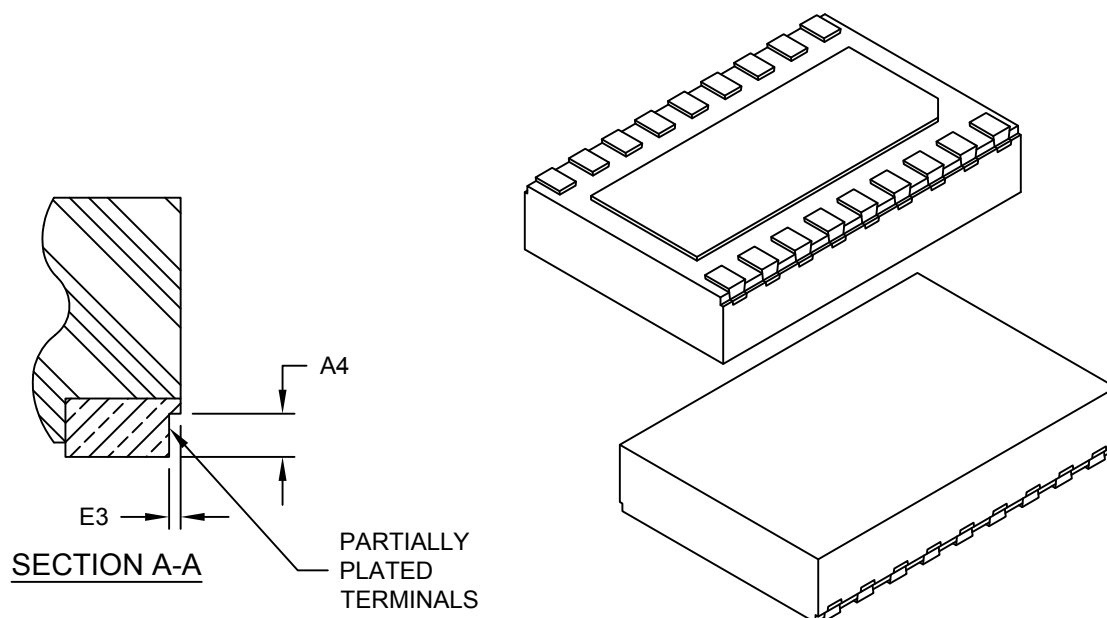


Microchip Technology Drawing C04-21458 Rev. B Sheet 1 of 2

Packaging Diagrams and Parameters

18-Lead Very Thin Dual Flatpack No-Lead Package (QQB) 4.5x3 mm Body (VDFN) With Stepped Wettable Flanks

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



		Units	MILLIMETERS		
Dimension Limits			MIN	NOM	MAX
Number of Terminals	N		18		
Pitch	e		0.50 BSC		
Overall Height	A		0.80	0.90	1.00
Standoff	A1		0.00	0.03	0.05
Terminal Thickness	A3		0.203 REF		
Overall Length	D		4.50 BSC		
Exposed Pad Length	D2		3.80	3.90	4.00
Overall Width	E		3.00 BSC		
Exposed Pad Width	E2		1.50	1.60	1.70
Terminal Width	b		0.20	0.25	0.30
Terminal Length	L		0.35	0.40	0.45
Terminal-to-Exposed-Pad	K		0.20	-	-
Wettable Flank Step	A4		0.10	-	0.19
Wettable Flank Step	E3		-	-	0.085

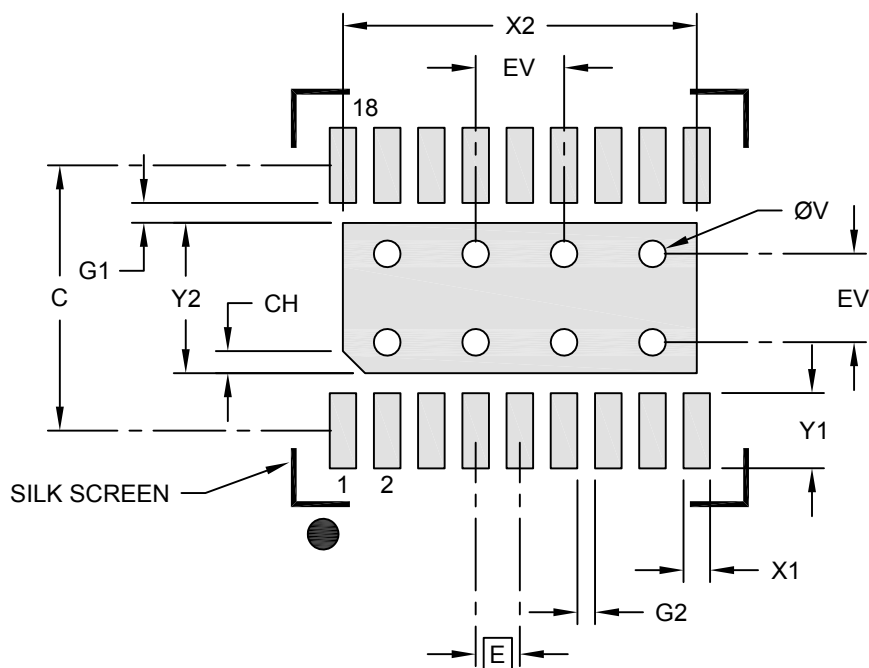
Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
 - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
 - REF: Reference Dimension, usually without tolerance, for information purposes only.

Land Pattern

18-Lead Very Thin Dual Flatpack No-Lead Package (QQB) 4.5x3 mm Body (VDFN) With Stepped Wettable Flanks

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC		
Optional Center Pad Width	X2			4.00
Optional Center Pad Length	Y2			1.70
Contact Pad Spacing	C		3.00	
Contact Pad Width (X20)	X1			0.30
Contact Pad Length (X20)	Y1			0.85
Pin 1 Index Chamfer	CH		0.25	
Contact Pad to Center Pad (X18)	G1	0.20		
Contact Pad to Contact Pad (X16)	G2	0.20		
Thermal Via Diameter	V		0.30	
Thermal Via Pitch	EV		1.00	

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process